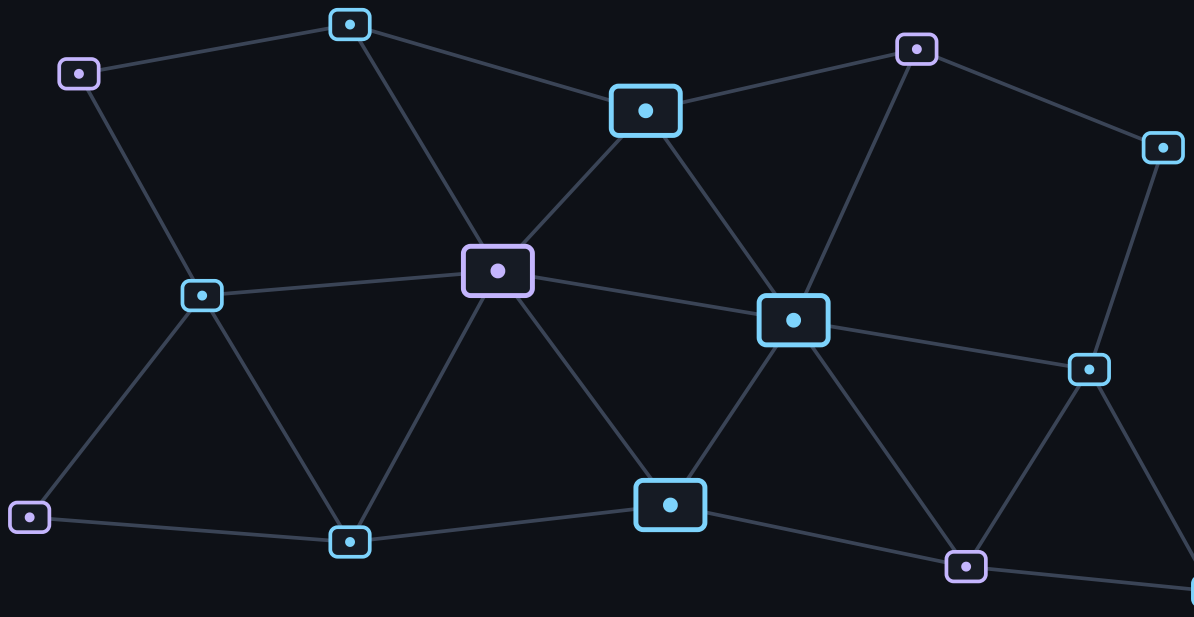
CMPN421 / CMPn421 · Computer Engineering

Distributed Computer Systems

Assignments 1 & 2 — Model Answers

Chapter 1 · Introduction to DCS | Chapter 2 · Moving Technologies and DCS Types



21
ASS#1 questions · Ch1

14
ASS#2 questions · Ch2

35
model answers

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2. Define data centers and mention its applications?
3. Mention Data Centers components?
4. Explain the key components of distributed computing systems using drawing?
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20. What are the benefits of a computer cluster?
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02 ASS#2 — Moving Technologies & Computing Types

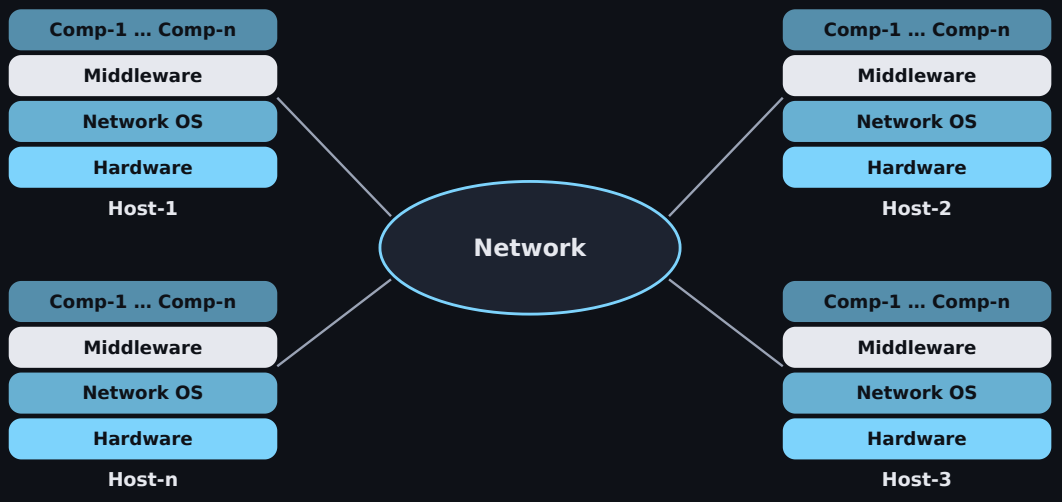
Ch2 · Moving Technologies and DCS Types — Q22–Q35

22. Explain the technologies that move the computer systems from centralized to distributed.
23. Mention the roles of microprocessor/minicomputer technology in the evolution of DCS?
24. Mention the roles of local area and wide area networks technology in the evolution of DCS?
25. Mention the roles of the PC technology in the evolution of DCS?
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27. Draw and explain Enslow model for distributed processing systems
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ASS#1 — DCS Introduction

Q1 ● متكرر Source: Ch1 — slide 43 (definition) · slides 4–5 (benefits) · slide 41 (comparison)

Define Distributed Computer System DCS? What are the pros and cons of Distributed system over centralized system?



Definition:

- A DCS is one in which components located at networked computers **communicate and coordinate their actions only by passing messages**.
- It requires **autonomous computers** interconnected through a network, each equipped with **DOS software**.
- DCS requires **transparency of distribution**: users are not aware that the system is distributed.

Pros over centralized:

1. **Scalability:** add more nodes as data grows, without a drop in performance.
2. **Fault tolerance:** if one node fails, the remaining nodes continue processing.
3. **Performance:** parallel processing of subtasks → reduced processing time.
4. **Large volumes of data:** data partitioning; each segment processed on a node.

Cons: more complex management (coordination of many nodes) · potentially higher latency due to network communication · security is harder.

الفكرة التعريف 4 نقط + 4 مميزات (Scalability, Fault tolerance, Performance, Large data) + العيوب من جدول المقارنة.

Q2

Source: Ch1 — slide 14

Define data centers and mention its applications?

- **Data Centers (DCs)** are the physical facilities that house the IT equipment and infrastructure necessary to **store, process, and distribute** vast and ever-growing volume of data.
- Housed equipment: servers, storage systems, network devices, cybersecurity devices.
- DCs provide the physical infrastructure and environment for the **secured, reliable and efficient** operation of IT systems.
- Designed with **multiple levels of redundancy** and physical + logical security.

Applications: cloud computing · data storage and backup · web and email hosting · big data processing · enterprise systems (ERP, banking).

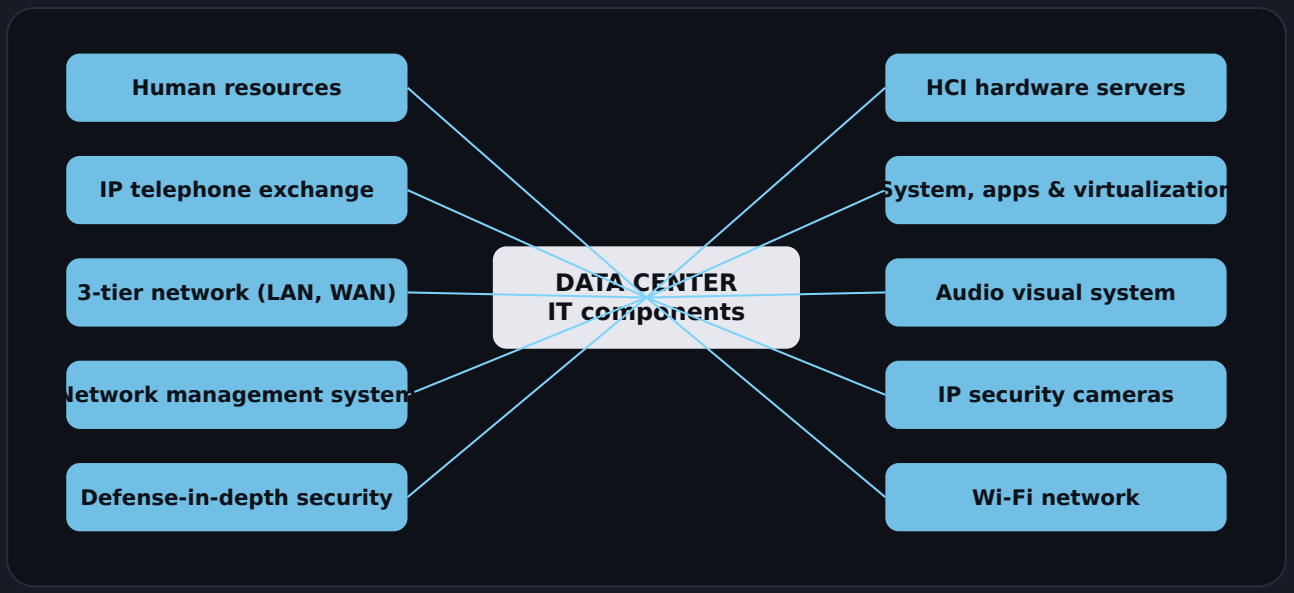
* التطبيقات مش مكتوبة في السلايدز؛ مضافة عشان السؤال بيطلبها.

الفكرة Store, process, distribute + redundancy + security.

Q3

Source: Ch1 — slide 15 (components) · slide 16 (IT components)

Mention Data Centers components?



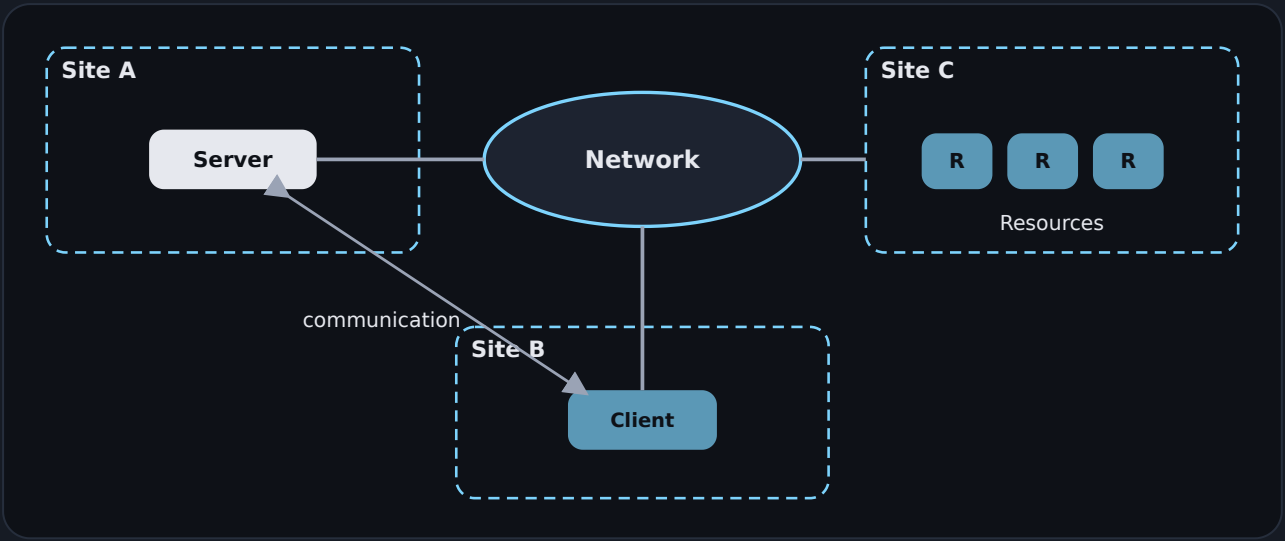
1. Human Resources
2. Physical Suitable Site
3. Racks and enclosures
4. Power Distribution Units (PDUs)
5. Cooling systems
6. Fire suppression systems
7. Security systems
8. Computer resources
9. Storage systems
10. Network equipment
11. Software and virtualization
12. Data Center Management

الفكرة component 12: ناس ومكان ← بنية (IT ← racks, PDU, cooling, fire, security) ← إدارة.

Q4 ● متكرر

Source: Ch1 — slide 38 · drawing: slide 89

Explain the key components of distributed computing systems using drawing?



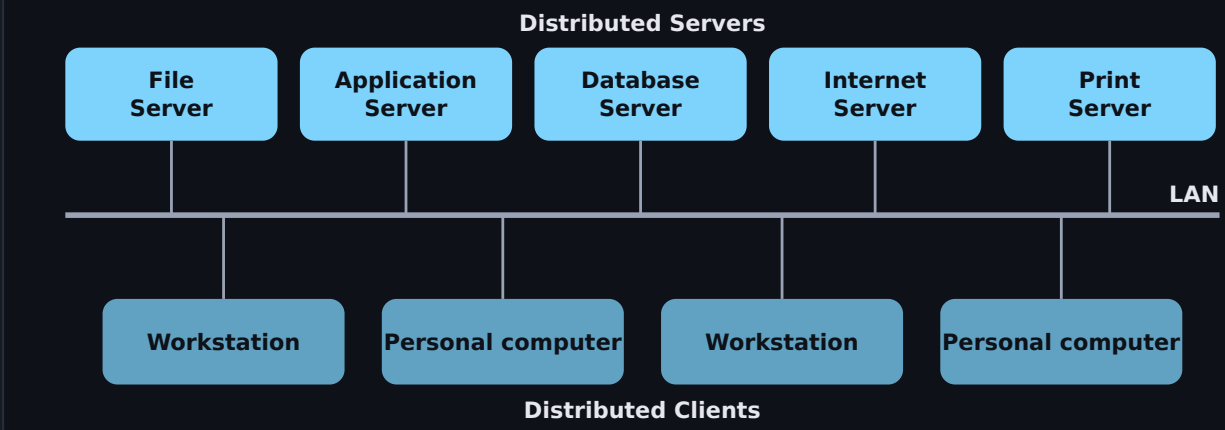
1. **Geographical Distribution:** nodes spread across different locations, communicating via LAN or the Internet.
2. **Resource Sharing:** nodes share processing power, storage, and data.
3. **Concurrency:** multiple nodes operate concurrently → better performance and throughput.
4. **Scalability:** easy to scale by adding more nodes.
5. **Fault Tolerance:** redundancy and replication keep the system operational if some nodes fail.
6. **Transparency:** complexity is hidden; users see a single entity.

الفكرة 6 خصائص + ارسم Site C – Site B (client) – Network – Site A (server) (resources).

Q5

Source: Ch1 — slide 37 · drawings: slides 89, 90

With the help of a diagram explain the working of distributed systems.



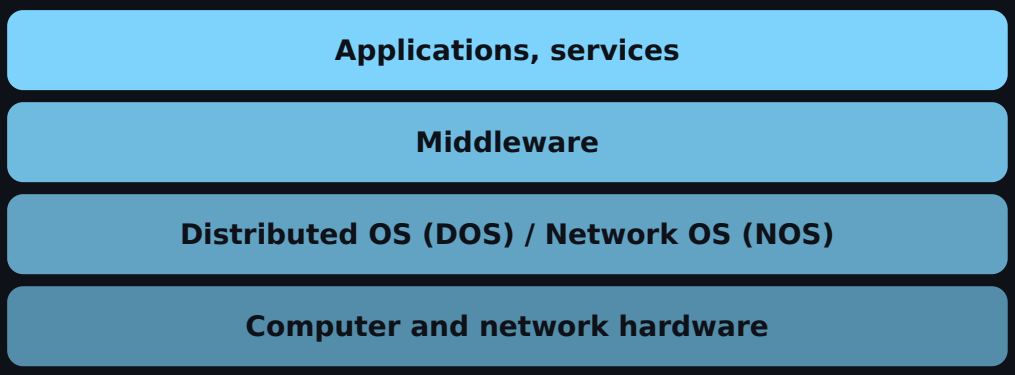
- Distributed systems are architectures where **multiple independent nodes work together to achieve a common goal**.
- Nodes **communicate and coordinate over a network**, appearing as a **single coherent system** to the end user.
- Distributed servers (file, application, database, internet, print) serve distributed clients (workstations, PCs) over the LAN.
- Aim: improve performance, reliability, scalability and resource sharing.

الفكرة Servers فوق + LAN + Clients تحت، وكلهم شغالين كأنهم system واحد.

Q6 ● متكرر

Source: Ch1 — slide 57 (layers) · slide 58 (NOS vs DOS)

Draw the DCS software architecture and compare between distributed operating system and network operating system?



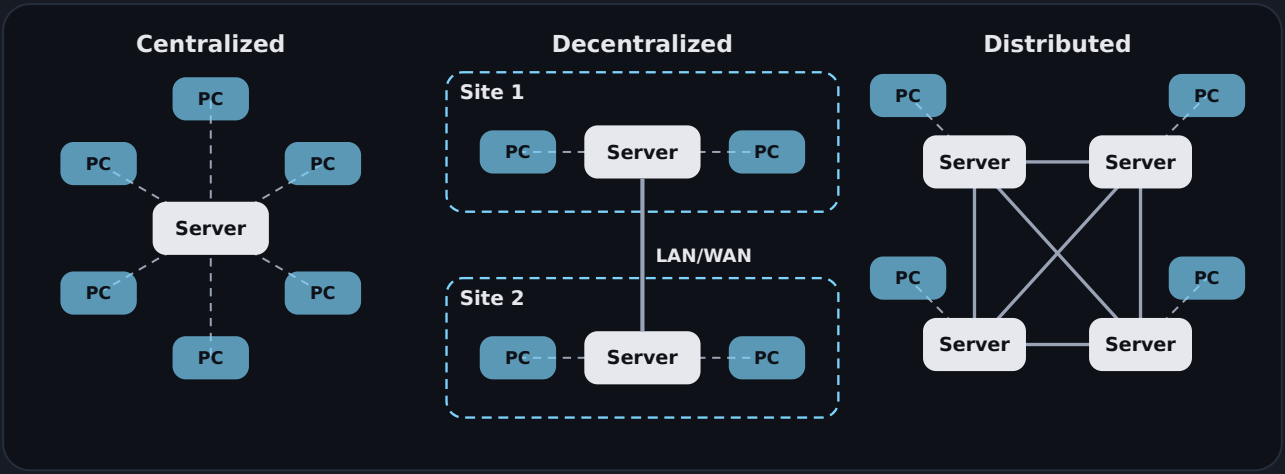
Network OS (NOS)	Distributed OS (DOS)
Supports networking; connects computers to share files, applications and printers	Coordinates connected nodes to operate like a **single large machine**
Each computer works self-contained, communicating through the NOS	DOS unifies all computers so they work together
MS Server, IBM AIX, Oracle Solaris, Linux Server, UNIX	MS Azure, Google File System (GFS), LOCUS
Resource sharing · centralized control · scalability · NOS security	Resource utilization · transparency · fault tolerance · scalability · DOS security

الفكرة كل layers: Applications ← Middleware ← DOS/NOS ← Hardware. NOS 4 جهاز لوحده، DOS الكل جهاز واحد.

Q7 متكرر

Source: Ch1 — slides 40, 41

Compare between centralized, decentralized, and distributed computing systems?



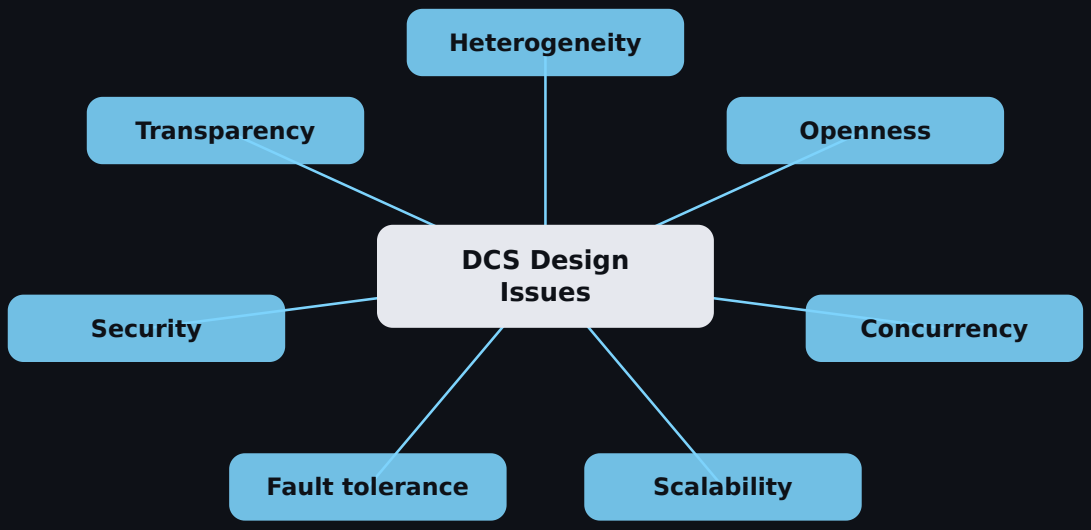
Aspect	Centralized	Decentralized	Distributed
Definition	Single central server controls all operations	Multiple nodes with independent control, no central authority	Interconnected nodes working together as a single system
Control	Single point of management	Each node operates independently	Shared; nodes collaborate
Single point of failure	High risk	Reduced risk	Reduced; fault tolerance and redundancy
Scalability	Limited, bottleneck	More scalable	Highly scalable
Resource utilization	Central server heavily utilized	Spread across nodes	Efficient sharing across nodes
Performance	Degrades with load	Improves with more nodes	High (parallel processing)
Management	Easier	More complex	Complex (coordination)
Latency	Lower	Varies with distance	Higher (network)

الفكرة احفظ الـ 8 Aspects وارسم الـ 3 رسومات.

Q8 ● متكرر

Source: Ch1 — slide 46 · Ch2 — slides 39–42

Identify the various challenges in the distributed system.



1. **Heterogeneity:** networks, hardware, OS, languages and developers differ → middleware.
2. **Openness:** adding new resource-sharing services → published interfaces.
3. **Security:** confidentiality, integrity, availability (CIA).
4. **Scalability:** remain efficient as users and resources grow.
5. **Failure handling:** partial failures → recovery and redundancy.
6. **Concurrency:** safe simultaneous access to shared resources.
7. **Transparency:** perceived as a single system.
8. **Resource sharing:** controlled sharing of hardware, data and services.

الفكرة 8 تحديات، كل واحد وجنبه حله.

Q9 متكرر

Source: Ch1 — slides 54, 55 · Section 2 (Q1)

Explain what is meant by (distribution) transparency, and give examples of different types of transparency.

Transparency ensures that the DCS is perceived as a **single entity** by the users or the application programmers rather than a collection of cooperating autonomous systems; the user is unaware where services are located.

Transparency	Description
Access	Hide differences in data representation and how a resource is accessed
Location	Hide where a resource is located
Migration	Hide that a resource may move to another location
Relocation	Hide that a resource may be moved to another location while in use
Replication	Hide that a resource is replicated
Concurrency	Hide that a resource may be shared by several competitive users
Failure	Hide the failure and recovery of a resource
Parallelism	Automatic use of parallelism without programming it explicitly

الفكرة 8 أنواع، كل نوع بيخبي حاجة واحدة.

Q10

Source: Ch2 — slide 26

Explain the differences between homogeneous and heterogenous systems?

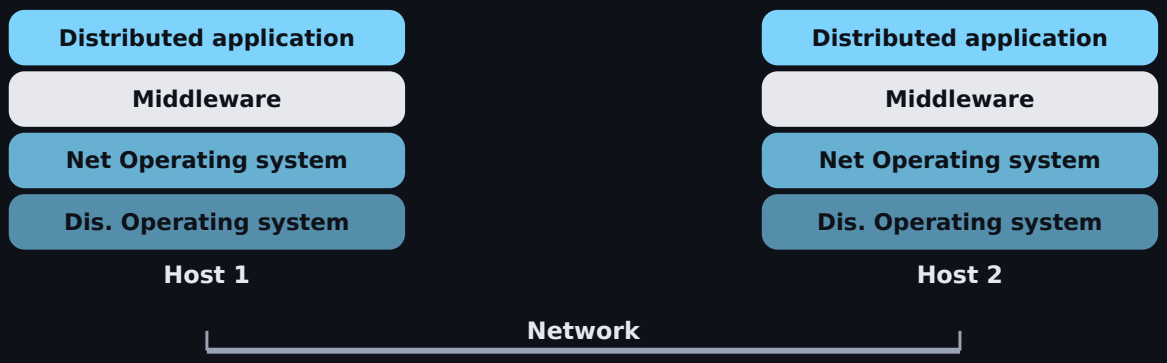
Subsystems	Homogeneous	Heterogeneous
Computer hardware	Same	Variant / Different
Operating systems	Same	Variant / Different
Programming languages	Same	Variant / Different
Implementations by developers/vendors	Same	Variant / Different
Networks	Same	Variant / Different
Design and management	Simple	Difficult
Growth	Simple	Difficult
Increased performance	Simple	Difficult

الفكرة Homogeneous = Same و Heterogeneous = Different و Simple. Difficult.

Q11 ● متكرر

Source: Ch1 — slide 60

List the functions of middleware.



1. Layer between (DOS and NOS) and distributed applications.
2. Hides complexity and heterogeneity of the distributed system.
3. Bridges the gap between low-level OS communications and programming language abstractions.
4. Provides common programming abstraction and infrastructure for distributed applications.

الفكرة 4 وظايف — أهمهم إنه بيخبي الـ heterogeneity.

Q12

Source: Ch1 — slide 50

Give examples for security threats and security protection.

Security objectives: **Confidentiality, Integrity, Availability (CIA)**. Encryption protects shared resources and keeps sensitive information secret when transmitted.

Security threats	Security protection (tools)
1) Threats to the processes	1. **Access rights** — who is allowed to perform the operations of an object
2) Threats to the servers	2. **Principal** — the authority of each session; a user or a process
3) Threats to the communication channel	3. **Encryption and authentication** — build secure channels
4) Denial of Service (DoS) attacks	

الفكرة CIA + 4 threats + 3 tools.

Q13

Source: Ch1 — slide 51

What is meant by scalability challenge and how to achieve it?

- The system should **remain efficient** even with a significant increase in the number of users and resources (10 or 100 nodes — performance shouldn’t vary).
- Scaling considers size, geography and management: accommodate more users and respond faster.

How to achieve:

1. Add more and/or faster processors.
2. Components should not need to be changed when the scale increases — **design components to be scalable**.
3. Load distribution across multiple nodes.

الفكرة Efficient مع زيادة المستخدمين ← add processors + scalable components.

Q14

Source: Ch2 — slide 42 · Ch1 — slide 38

What is meant by resources sharing challenge and how to achieve it?

- Resources (hardware like printers, data like files, services like search engines) are physically encapsulated within computers and accessed from others only by communication.
- Decentralized sharing raises challenges: system-wide coordination, security, user authenticity and provider policy heterogeneity.

How to achieve: each resource is managed by a program that offers a **communication interface**, so the resource is accessed and updated **reliably and consistently**.

الفكرة الحل: communication interface لكل resource.

Q15

Source: Ch1 — slide 49 · Ch2 — slide 40

What is meant by openness challenge and how to achieve it?

Openness is determined by the degree to which **new resource-sharing services** can be added and made available to users.

How to achieve:

1. Key interfaces are **published** (specification and documentation available).
2. A **uniform communication mechanism** and published interface for access to shared resources.
3. New components integrated with existing components (heterogeneous hardware and software).
4. Differences in data representation between vendors are resolved.

الفكرة Openness = published interfaces + uniform communication.

Q16

Source: Ch1 — slide 52

What is meant by fault tolerance challenge and how to achieve it?

- Faults in hardware or software may produce incorrect results or stop before completing the computation.
- Failure handling is difficult because the failure is **partial**: some components fail while others continue to function.
- Distributed systems must maintain **availability** even at low levels of hardware/software/network reliability.

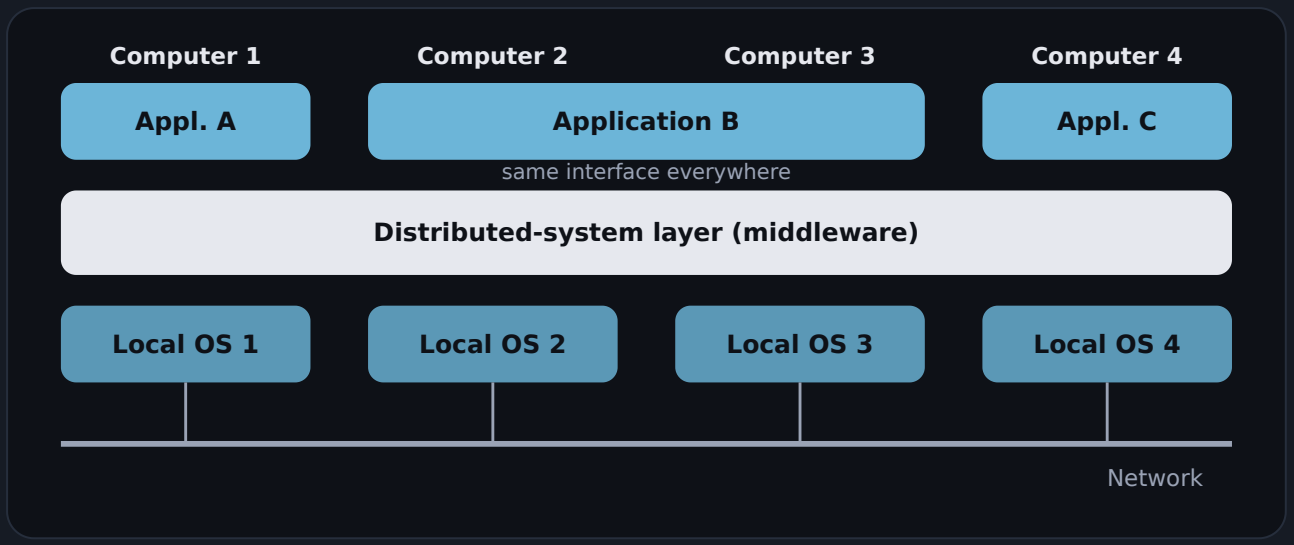
How to achieve: recovery and **redundancy**.

الفكرة Failure partial ← recovery + redundancy.

Q17 ● متكرر

Source: Ch1 — slides 47, 48, 60

Explain and List the functions of middleware.



Middleware is a set of services that enables applications and end-users to interact across a heterogeneous DCS by **masking the heterogeneity** of the underlying networks, hardware, OS and programming languages; it offers each application the same interface.

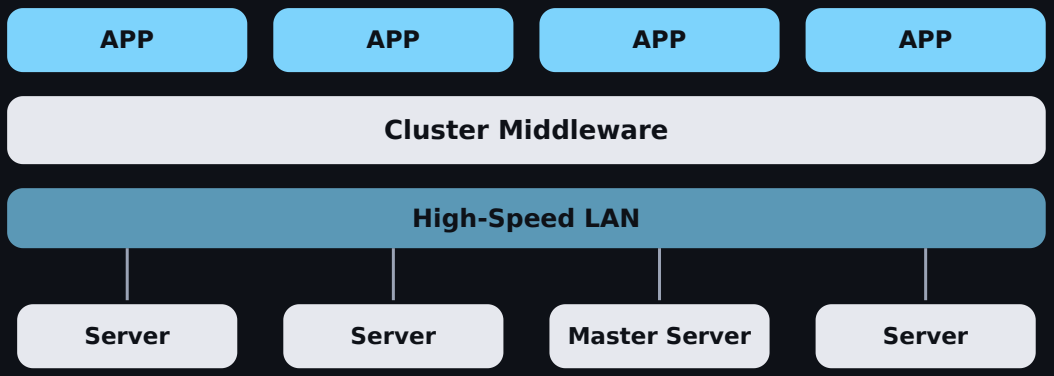
1. Layer between (DOS and NOS) and distributed applications.
2. Hides complexity and heterogeneity.
3. Bridges low-level OS communications and programming language abstractions.
4. Common programming abstraction and infrastructure.

الفكرة تعريف ((masking heterogeneity + نفس الـ 4 وظايف.

Q18

Source: Ch1 — slides 62, 64

Explain and List the functions of clustering computing.



- A cluster is a group of linked computers working together so closely that they form a **single computer**, connected through **fast LANs**.
- A large task is broken into discrete parts distributed across computers and processed simultaneously.

Functions: improve **performance and/or availability** more cost-effectively than one computer · parallel processing · redundancy (failover) · load balancing.

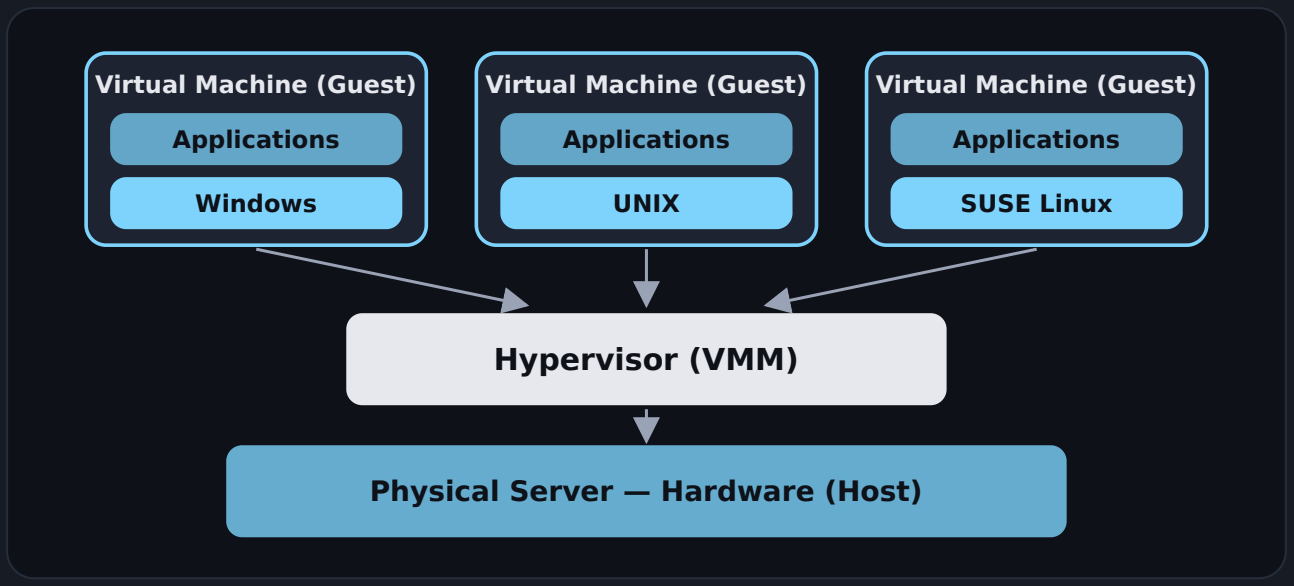
Cluster consists of: nodes (master + computing) · network · OSs · cluster middleware.

الفكرة Cluster = كذا كمبيوتر على LAN سريعة كأنهم كمبيوتر واحد.

Q19

Source: Ch1 — slides 67–69, 71

Explain and List the functions of virtualization.



- Virtualization partitions the resources of a single platform into **multiple segregated, virtualized execution environments**, allowing multiple OSs on the same hardware.
- **Guest** = each environment · **Host** = the platform · **Hypervisor (VMM)** = the bridge between them.
- **CPU virtualization:** the hypervisor preempts an OS that consumes all the CPU. **Memory virtualization:** each OS has a virtual address space mapped to physical memory.

Functions / advantages:

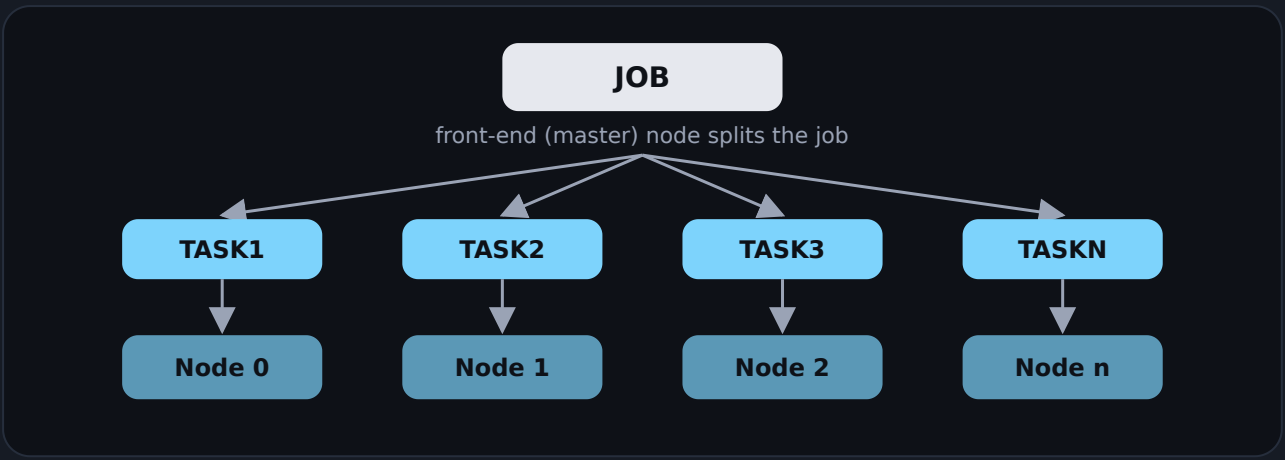
1. **Multiple secure environments** (sandbox).
2. **Mixed OS environment.**
3. **Better system utilization.**
4. **Failure isolation.**

الفكرة Guest / Host / Hypervisor + 4 مميزات.

Q20

Source: Ch1 — slides 62, 66

What are the benefits of a computer cluster?



1. **Performance:** a job is split into tasks processed simultaneously → higher speed.
2. **Availability:** redundancy; no single point of failure.
3. **Cost-effective:** cheaper than a single computer of comparable speed or availability.
4. **Scalable parallel and distributed computing** in science and business.
5. **Cloud services:** high-performance computing, mass storage (data centers), web search.

الفكرة Performance + Availability + Cost + Cloud services.

Q21

Source: Ch1 — slide 63

What are the types of cluster computing?

1. **High availability (HA) clusters (Linux):** for mission-critical applications; failover clusters provide redundancy and **eliminate single point of failure**.
2. **Network load balancing clusters:** distribute a workload over multiple backend nodes with redundant load-balancing front ends; all servers (web, mail, FTP) process requests.
3. **Science clusters:** a **master** computer breaks the problem into pieces and sends a piece to each **node**, sending more as nodes finish → cheap powerful supercomputer.

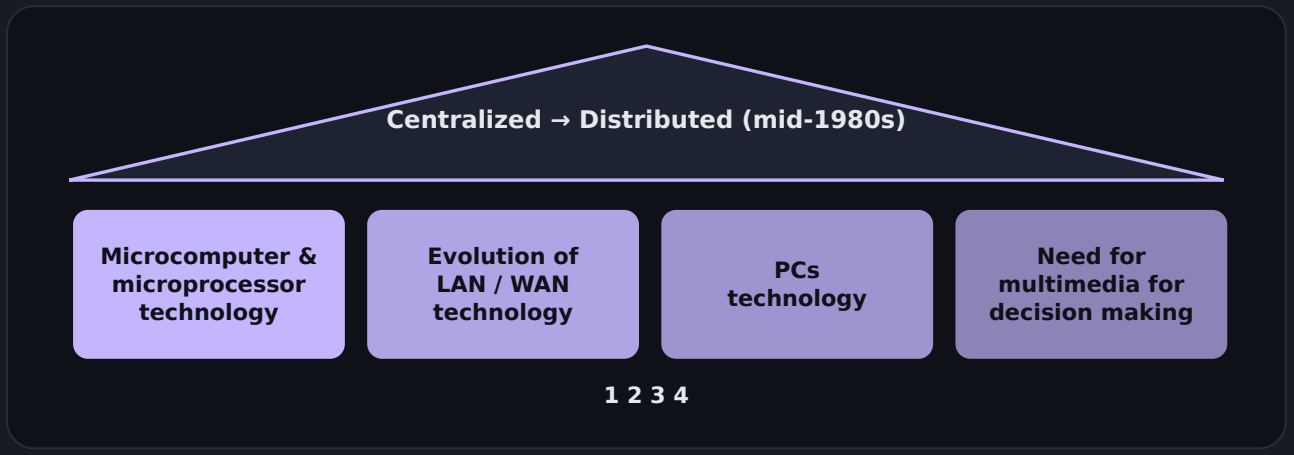
الفكرة 3 أنواع: HA – Load balancing – Science.

ASS#2 — Moving Technologies & Computing Types

Q22 ● متكرر

Source: Ch2 — slides 3, 15 · Section 2 (Q3)

Explain the technologies that move the computer systems from centralized to distributed.



1. **Microcomputer and microprocessor technology:** 8 → 64-bit and multicore CPUs; mainframe power for 1/1000th of the price.
2. **Evolution of LAN technology:** high-speed LANs and WANs connect thousands to millions of machines.
3. **PCs technology:** miniaturization — smartphones, plug and nano computers.
4. **Need for multimedia for decision making:** PCs and powerful servers replace text terminals.

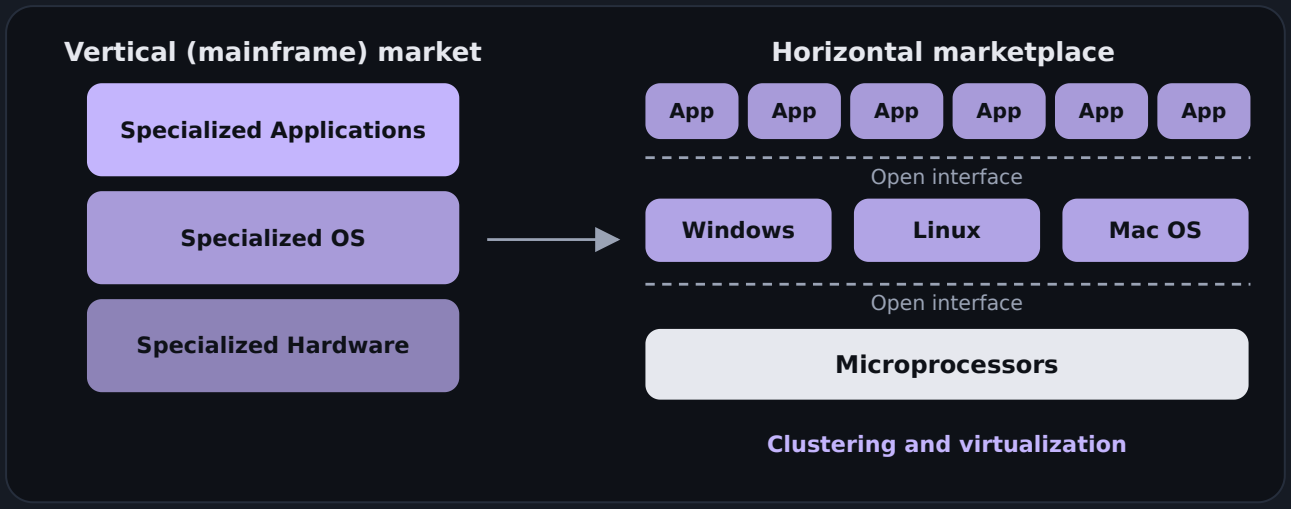
Feature	Notes
Technological changes	VLSI favors multiple low-performance processors; communication costs fell; PCs replace terminals.
User needs	Computing power where needed; managers need multimedia data for decisions.

الفكرة 4 تكنولوجيات: Microprocessor, LAN/WAN, PCs, Multimedia.

Q23

Source: Ch2 — slides 4–7

Mention the roles of microprocessor/minicomputer technology in the evolution of DCS?



- **Powerful microprocessors:** 8-bit → 16, 32, 64-bit and multicore CPUs → challenge of exploiting clustering and parallelism.
- Current machines have the power of mainframes of 30–40 years ago for **1/1000th of the price**.
- The **vertical mainframe market** became a **horizontal marketplace** with open interfaces (Intel x86, Motorola 68000, BSD Unix, Linux).
- **HCI:** software-defined infrastructure; a cluster of at least 3 processors with virtualized computing.

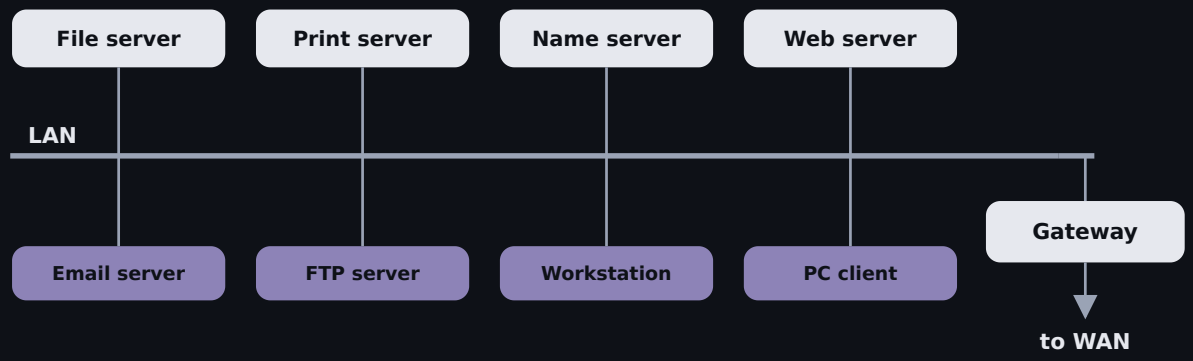
Categories: PC · handheld · Internet appliance · mid-range server · mainframe · supercomputer.

الفكرة قوة أكبر بسعر أقل + السوق بقى horizontal + مثال HCI.

Q24

Source: Ch2 — slides 8, 9

Mention the roles of local area and wide area networks technology in the evolution of DCS?



- Invention of **high-speed LAN and WAN**: LANs connect thousands of machines (up to 10^{12} bps); WANs connect millions (10^{4}–10^{9} bps).
- **File servers** provide data storage; **name servers** map machine names to IP addresses and user/group names to IDs; **print servers** control shared printers.
- A **gateway** connects the LAN to the WAN; LANs connected together form **Intranets**.

الفكرة LAN = file + name + print servers + gateway.

Q25

Source: Ch2 — slides 10, 12, 15

Mention the roles of the PC technology in the evolution of DCS?

- **Miniaturization:** the smartphone is a full-fledged computer (sensors, memory, powerful CPU).
- **Plug and Nano computers** offer near-desktop performance.
- PCs and mobiles are powerful, low-cost, multifunction devices that **replace terminals**.
- Decision makers need multimedia → text terminals replaced by PCs and powerful servers.

الفكرة الـ PC بقى بديل للـ terminal.

Q26

Source: Ch2 — slides 6, 11

Mention the different applications/configurations of the PC in the evolution of DCS?

- **Tower model** — tall, on the floor
- **All-in-one** — monitor + system unit
- **Workstation** — intense calculation and graphics
- **Notebook** — portable, on the lap
- **Handheld / PDA** — fits in the hand
- **Internet appliance** — connects to the Internet

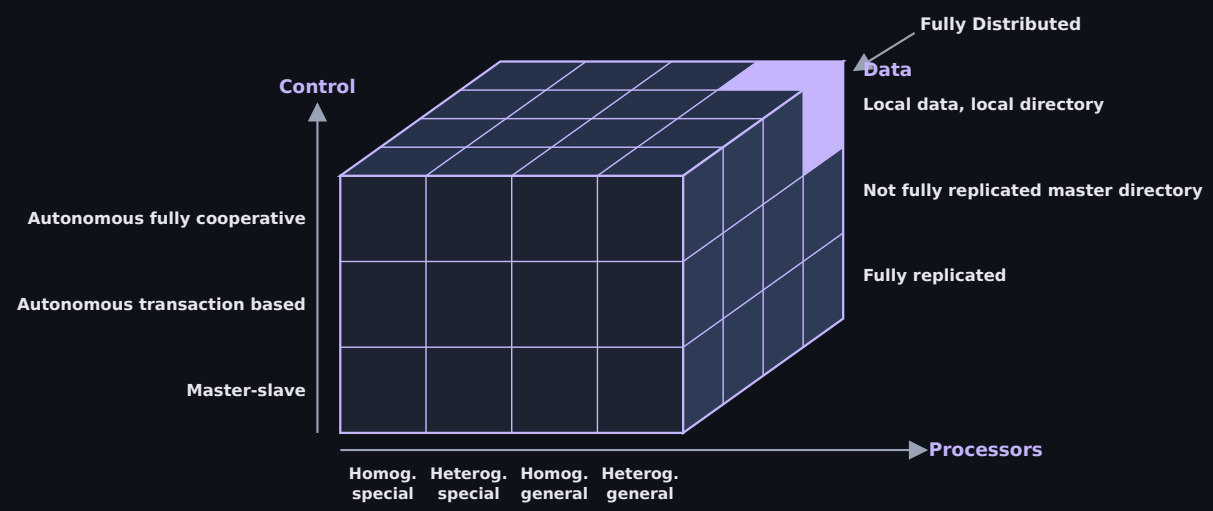
PC in a DCS: dumb terminal (emulation mode) · standalone PC · diskless PC · thin client · thick client.

الفكرة + Desktop (tower, all-in-one, workstation) + notebook + handheld + internet appliance.

Q27 ● متكرر

Source: Ch2 — slides 22–25 · Section 1 (Q4)

Draw and explain Enslow model for distributed processing systems



Distributed system = distributed hardware + distributed control + distributed data. A system is distributed if all three reach a certain degree of decentralization; **fully distributed** = fully distributed in all three.

Control:

1. Single fixed control point
2. Single dynamic control point
3. Fixed master/slave
4. Dynamic master/slave
5. Multiple homogeneous
6. Multiple heterogeneous

Data:

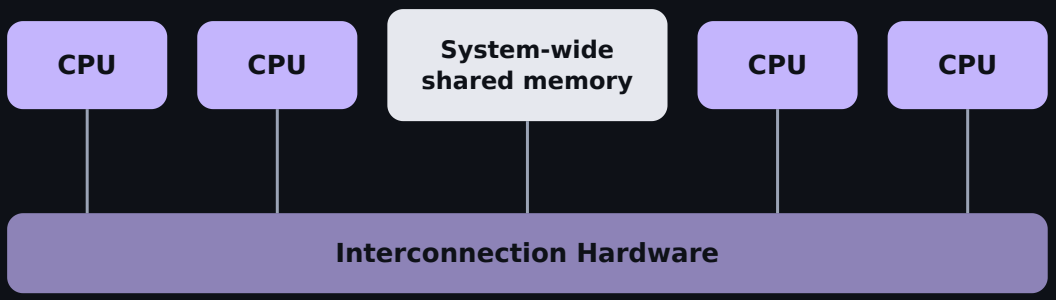
1. Centralized database
2. Distributed files, central directory
3. Replicated database
4. Partitioned, master keeps all files
5. Partitioned, master keeps directory
6. Partitioned, no master

الفكرة مكعب بـ 3 محاور (Processors, Control, Data)، والركن = fully distributed.

Q28

Source: Ch2 — slide 28

Explain Tightly Coupled Multiprocessor System?



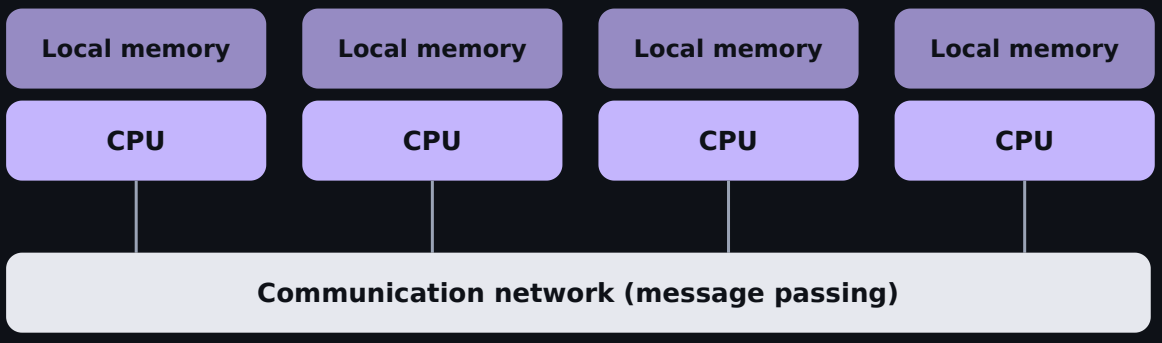
- Tightly coupled systems are **multiprocessor systems** sharing addressing to memory and devices using a **single OS**.
- Called parallel processing or **shared memory** systems: one system-wide primary memory shared by all processors.
- If a processor writes 100 to location x, any other processor reading x gets 100.
- The number of processors is small, limited by the bandwidth of the shared memory.

الفكرة Shared memory + single OS.

Q29

Source: Ch2 — slide 29

Explain Loosely Coupled Multiprocessor System?



- Comes very close to distributed processing: **multiple complete systems**.
- Connected over I/O pathways; they **do not share memory or operating systems**.
- They may be functionally dependent and act as a single node.

الفكرة Private memory، مفيش shared memory ولا OS.

Q30

Source: Ch2 — slide 30

Differentiate between Loosely Coupled and Tightly Coupled Multiprocessor Systems

Loosely Coupled	Tightly Coupled
Distributed memory	Shared memory
Low data rate	High data rate
Memory conflicts don’t take place	Memory conflicts
Low degree of interaction between tasks	High degree of interaction between tasks
Direct connection between processor and I/O devices	IOPIN connects processor and I/O devices
Applications: DCSs	Applications: parallel processing

الفكرة كل سطر عكس اللي قصاده.

Q31

Source: Ch2 — slides 31, 32

Explain networking distributed Multiprocessor System?

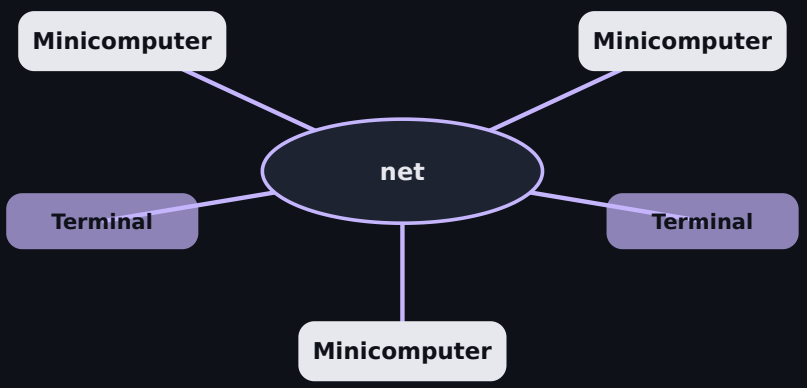
- Loosely coupled systems using networks are referred to as **DCS**: processors do not share memory; each has its own **local memory**.
- A write of 100 to x changes only the local memory of that processor.
- All communication is by **passing messages** across the interconnection network.
- Processors can be far apart → DCS are freely expandable with an almost unlimited number of processors.
- A processor and its resources are called a **node / site / machine**; its own resources are local, others are remote.

الفكرة .Private memory + message passing = DCS

Q32

Source: Ch2 — slide 34

Explain minicomputer distributed Multiprocessor System?



- A simple extension of the **centralized time-sharing system**.
- A few minicomputers on a network, each with multiple users and interactive terminals.
- Each user is logged on one minicomputer with **remote access** to the others. Example: **early Internet**.

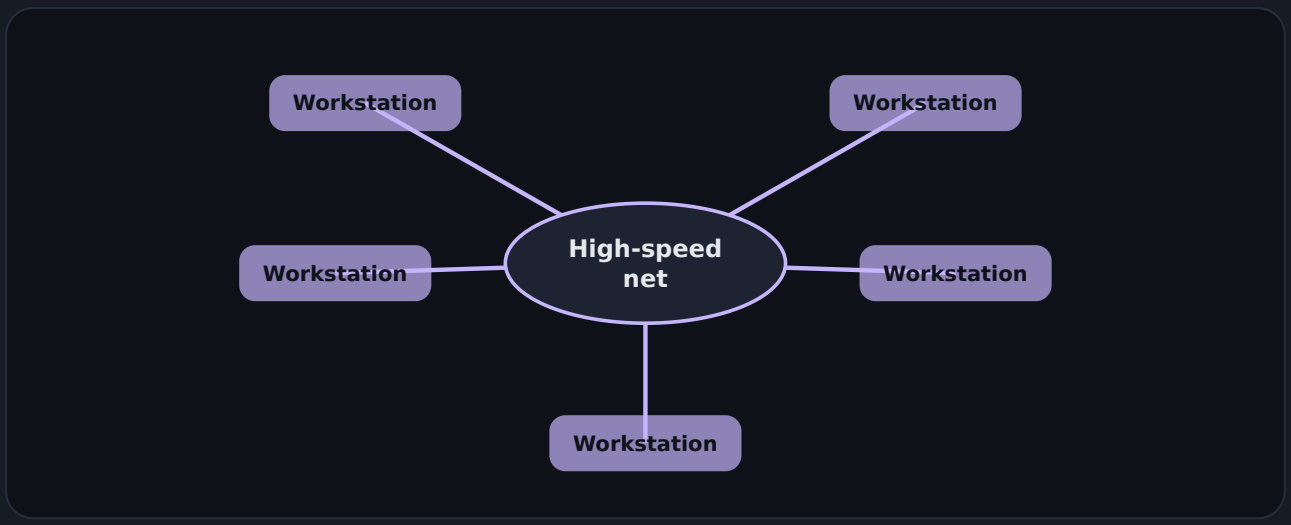
Drawbacks: tasks processed by a single minicomputer · very expensive.

الفكرة Minicomputers + terminals، مثال early Internet، وغالي.

Q33

Source: Ch2 — slide 35

Explain workstation distributed Multiprocessor System?



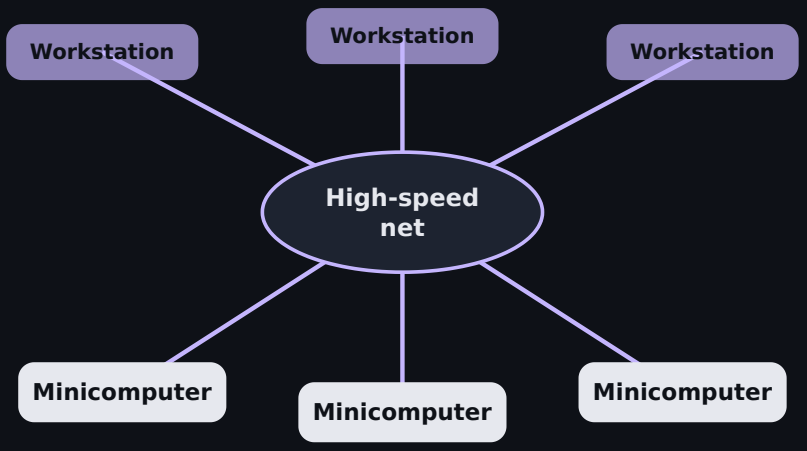
- A network of powerful PCs, each with its **own hard disk** and local file system, on a **high-speed LAN**.
- At any time many workstations are **idle**, wasting CPU time.
- Idle workstations process jobs of users logged onto other workstations. Example: **Sprite system, Xerox PARC**.

الفكرة استغلال الـ workstations الـ idle.

Q34

Source: Ch2 — slide 36

Explain workstation server distributed Multiprocessor System?



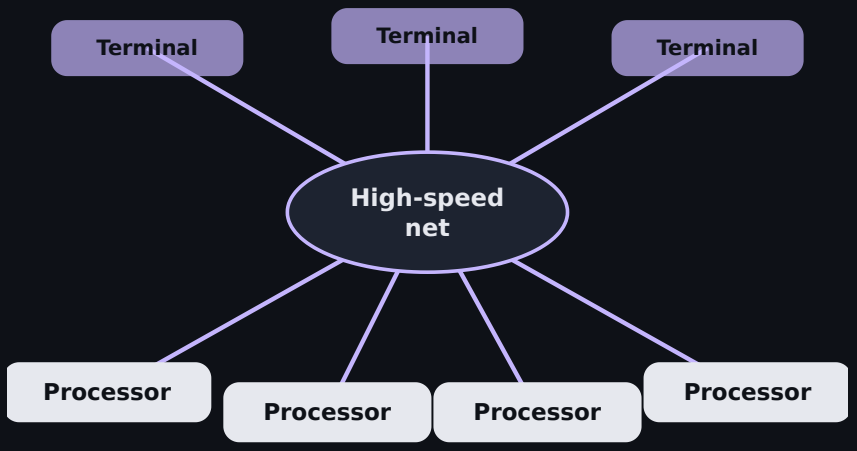
- Multiple workstations coupled with **powerful servers** (file systems, databases).
- A user logs onto his own workstation; normal computation is done locally.
- Requests for special services are sent to a **server** that performs them and returns the result; processes need not migrate to the server.

الفكرة الشغل العادي local، والخدمات الخاصة على الـ server.

Q35

Source: Ch2 — slide 37

Explain Multiprocessor pool distributed Systems?



- Most of the time a user needs no computing power, but once in a while needs a **very large amount** for a short time.
- Processors are **pooled** and shared by users as needed; each has its own memory.
- No home machine; the user does not log onto any machine → better utilization and flexibility. Example: **Cambridge DCS**.

الفكرة Pool مشترك + مفيش home machine.